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Revealing Hidden Jump Strategies Beyond Conventional Force-Plate Metrics

Analysis Report

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Data: VALD ForceDecks countermovement jumps, male football athletes, high-performance centre, Thane, Mumbai (805 trials, 212 athletes)
Main analysis: lowest-centre-of-mass landing window · Sensitivity analysis: 1-s landing window

Executive summary

Objective. Force plates summarise a countermovement jump (CMJ) with a handful of vendor metrics, and athletes with similar scores are usually treated as equivalent. This report asks whether male football players who reach the same jump height still produce different force–time curves, and how much of that curve-shape variation the conventional metrics capture.

Data and approach. 805 CMJ trials from 212 athletes (after quality control of 833 trials from 215 athletes) recorded on dual ForceDecks plates were normalised to body weight. Take-off and landing phases were each time-normalised to 0–100%, joined end to end with flight excluded, and analysed with Principal Component Analysis (PCA).

Key findings

1. **A few shape patterns describe most of the curve.** Four principal components explain 74.8% of curve variance (PC1 37.3%, PC2 16.0%, PC3 12.6%, PC4 8.9%).
2. **The landing phase carries the dominant variation.** More than 99% of each retained component's loading lies in the landing phase; PC1 is most strongly related to landing peak force ($r = 0.39$), not to take-off timing ($r = -0.04$) or countermovement depth ($r = 0.01$).
3. **Matching on jump height leaves curve spread largely intact.** Among 82 trials matched on jump height (27.1 ± 0.7 cm), the SD of PC1 is 6.45 versus 7.28 in the full sample (ratio 0.89; athlete-bootstrap 95% CI 0.70–1.06); total curve variance is unchanged (SD ratio 1.01). Matching on relative peak force gives a whole-curve ratio of 1.00.
4. **Conventional metrics explain little of the curve shape.** The four vendor metrics plus age and body mass explain 2.7–13.1% of the variance of each retained component, leaving 86.9–97.3% unexplained; with athlete-grouped cross-validation the out-of-sample figure is 0.3–11.8%.
5. **The conclusion depends on how the landing phase is defined.** With a fixed 1-s landing window the take-off timing component becomes 68.3% predictable from the metrics, so the "largely unexplained" conclusion holds for landing-dominated components only. The numbers also differ modestly from the submitted abstract (Section 6).

1. Objective and research questions

Routine force-plate testing in high-performance centres produces thousands of complete force–time recordings, but decisions are usually based on a few vendor-computed metrics such as jump height, peak force and phase impulses (Cormie et al., 2009; McMahon et al., 2018). Two athletes with the same jump height may reach it with different movement strategies, and those differences sit in the shape of the curve rather than in its summary values (Richter et al., 2014b). How force is produced within individual sub-phases of the jump is itself related to performance (Harry et al., 2018).

The study addresses one plain question: *once jump height is fixed, do players still move differently, and do the standard vendor metrics capture the shape of the curve?* It is broken into three research questions:

1. **RQ1 – Structure.** What are the main patterns of variation in full CMJ force–time curves (take-off and landing)?
2. **RQ2 – Hidden variation.** Among trials matched on jump height or on relative peak force, is the spread of curve shapes smaller than in the full sample?
3. **RQ3 – Redundancy.** How much of the variation in curve shape is explained by the four conventional vendor metrics, age and body mass?

2. Data

Source. CMJ tests conducted at a high-performance centre in Thane, Mumbai, on VALD ForceDecks dual force plates. Raw vertical-force recordings for the left and right plates and the vendor's per-trial results were retrieved through the ForceDecks external API. Recordings were sampled at 1000 Hz; 60 of the 833 trials were recorded at 500 Hz, which does not affect the analysis because every phase is resampled to a common grid.

Population. Male football athletes. Each trial is linked to four vendor metrics – jump height (flight-time method), concentric peak force, eccentric braking impulse and concentric impulse over the first 100 ms – and to the athlete's age and body mass at the test. Athletes are identified only by pseudonymous codes; the data contain no names, dates of birth, test dates or vendor IDs.

Quality control. Rules R1–R4 were set before any results were examined; R5 is required by the lowest-centre-of-mass landing definition (Table 1). Body mass must be valid because every curve is divided by body weight; the quiet-stance check catches wrongly entered body mass or an athlete who was not still before the jump; the flight check removes trials where a foot stayed on a plate; the last rule removes trials in which the end of the landing phase (Section 3.3) could not be found.

Table 1. Quality-control flow (main analysis).

Step	Trials	Athletes
All male football CMJ trials (VALD API)	833	215
R1: valid VALD body mass	828	212
R2: quiet-stance mass within 5% of body mass	813	212
R3: no force on plate during flight	811	212
R4: complete metrics and age	811	212
R5: landing end found	805	212

Table 2. Descriptive statistics of the analysed sample (805 trials, 212 athletes). Age and body mass are per athlete at the first analysed test; other rows are per trial.

Variable	n	Mean ± SD	Median (IQR)	Range
Age at first test (years), per athlete	212	16.01 ± 5.94	14.10 (12.00–16.80)	8.80–37.90
Body mass at first test (kg), per athlete	212	53.31 ± 15.90	51.73 (40.87–64.03)	25.03–95.97
Jump height, flight time (cm)	805	28.19 ± 8.43	26.96 (21.62–34.69)	8.54–51.47
Concentric peak force (N)	805	1269.26 ± 420.74	1227.01 (923.47–1582.99)	473.52–3073.22
Concentric peak force (× BW)	805	2.35 ± 0.38	2.29 (2.11–2.51)	1.65–4.70
Eccentric braking impulse (N s)	805	27.80 ± 15.43	24.60 (16.59–35.70)	0.00–95.18
Concentric impulse 100 ms (N s)	805	62.24 ± 26.66	58.70 (41.76–79.14)	1.51–187.93
Peak landing force (× BW)	805	5.39 ± 1.97	4.97 (3.88–6.67)	2.19–14.88
Contraction time (s)	805	0.78 ± 0.19	0.77 (0.66–0.90)	0.31–1.48
Countermovement depth (cm)	805	-26.55 ± 9.90	-27.13 (-33.55–-20.71)	-52.39–0.00
Trials per athlete	212	3.80 ± 2.28	3.00 (2.00–5.00)	1.00–16.00

Countermovement depth is the vendor's centre-of-mass displacement, negative downwards. BW = body weight.

3. Methodology

3.1 Body-weight normalisation

Total vertical force (left + right plate) was divided by the athlete's body weight (body mass × 9.81 m/s^2). Heavier athletes produce larger forces simply because they are heavier; expressing force in multiples of body weight puts all athletes on the same scale, so that the analysis compares how athletes move rather than how much they weigh. A value of 1 means the plate is carrying exactly the athlete's weight.

3.2 Separate time-normalisation of take-off and landing

Trials last different lengths of time. If each whole trial were stretched to a single 0–100% scale, the same event (for example take-off) would fall at a different percentage in every trial, and PCA would mistake differences in timing for differences in shape. The trial was therefore cut at the vendor's event times – movement onset, take-off and touch-down (Owen et al., 2014; McMahon et al., 2018; Harry et al., 2020) – and each phase was resampled separately to 101 points (0–100%). This aligns take-off at the end of the first block and touch-down at the start of the second block in every trial, a simple landmark alignment (Ramsay & Silverman, 2005; Richter et al., 2014a).

3.3 Exclusion of flight and definition of landing

During flight the force is zero, so this part of the curve carries no shape information; its only feature is its duration, which determines flight-time jump height (Linthorne, 2001) – a metric that is already measured. Including it would let the analysis reconstruct jump height directly from the curve. The take-off block (onset → take-off) and the landing block were therefore joined end to end without the flight phase, giving 202 points per trial.

The landing phase starts at touch-down and ends at the **lowest point of the centre of mass**, the moment the downward velocity after touch-down returns to zero, which marks the end of the impact-absorption part of the landing. Landing characteristics of the CMJ are known to vary with how the jump is performed (Hite et al., 2025), which is why the landing is analysed here rather than discarded. Velocity at touch-down was set to −g × flight time / 2 and updated by integrating (force − body weight) / body mass. A

fixed 1-s window after touch-down (the vendor's trial end) was analysed as a sensitivity check (Section 5.1).

3.4 Principal Component Analysis

PCA (Jolliffe, 2002; Jolliffe & Cadima, 2016) was applied to the 202-point curves (centred, not scaled, since every point is in the same unit). Each principal component (PC) is a pattern of deviation from the mean curve; each trial gets a score on each PC saying how strongly it shows that pattern. Four components were retained, as specified in the abstract. Each PC is interpreted by plotting the mean curve ± 2 SD of the PC (Warmenhoven et al., 2019) and by correlating scores with interpretable descriptors (landing peak force, contraction time, countermovement depth and the vendor metrics).

3.5 Matched-band dispersion

To test whether equal scores mean equal curves, trials between the 45th and 55th percentile of jump height were selected, and separately of concentric peak force relative to body weight. The SD of each PC score inside the band was divided by its SD in the full sample. A ratio near 1 means that fixing the metric does not make the curves more alike; a ratio well below 1 means the metric already determines much of the shape. The same ratio was computed for the whole curve (square root of the summed point-wise variance).

3.6 Regression and uncertainty

Each retained PC score was regressed on the four vendor metrics, age and body mass by ordinary least squares. R^2 is the share of the component's variance explained; $1 - R^2$ is the share left unexplained. Because athletes contribute several trials, uncertainty in the band SD ratios was estimated with an athlete-level bootstrap (Efron & Tibshirani, 1993): athletes, with all their trials, were resampled 2,000 times (seed 2026), and the 2.5th and 97.5th percentiles are reported. Further robustness checks are described in Section 5.

4. Results

4.1 Mean curve and variance explained

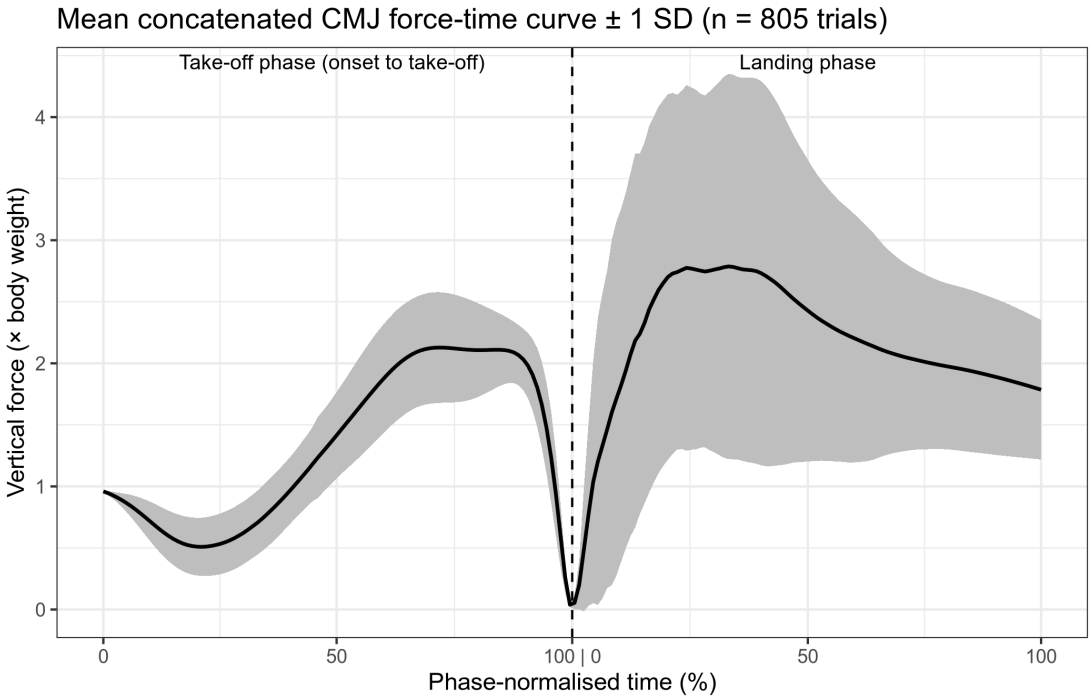


Figure 1. Mean body-weight-normalised force–time curve ± 1 SD (805 trials). Left: take-off phase (movement onset to take-off). Right: landing phase (touch-down to lowest centre of mass). Flight is excluded.

The first component explains 37.3% of the variance and the first four together 74.8% (Table 3, Figure 2). Spread is much larger in the landing block than in the take-off block (Figure 1): landing forces reach several times body weight and differ widely between trials.

Table 3. Variance explained by the first six components.

Component	SD of score	Variance (%)	Cumulative (%)
PC1	7.28	37.3	37.3
PC2	4.77	16.0	53.3
PC3	4.22	12.6	65.9
PC4	3.56	8.9	74.8
PC5	2.88	5.8	80.6
PC6	2.31	3.7	84.4

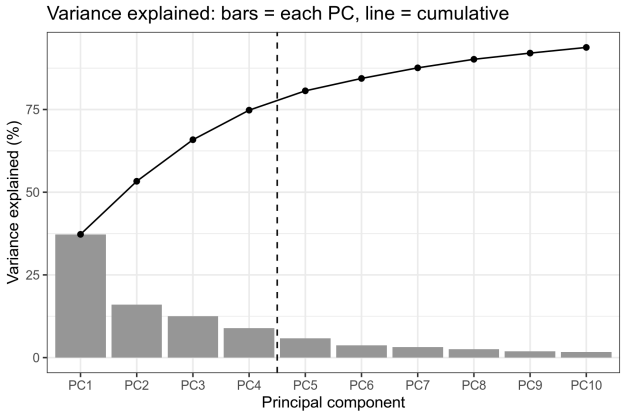


Figure 2. Scree plot: bars show each component; the line is cumulative. The dashed line marks the four retained components.

4.2 What the components represent

All four retained components act almost entirely in the landing phase (Table 4). Figure 3 shows the mean curve with each component added and subtracted (± 2 SD).

Table 4. Where each retained component acts: share of the squared loading falling in the take-off and landing blocks, and the point with the largest loading.

Component	Variance (%)	Take-off share of loading (%)	Landing share of loading (%)	Largest loading at
PC1	37.3	0.3	99.7	Landing 40%
PC2	16.0	0.1	99.9	Landing 26%
PC3	12.6	0.5	99.5	Landing 16%
PC4	8.9	0.7	99.3	Landing 9%

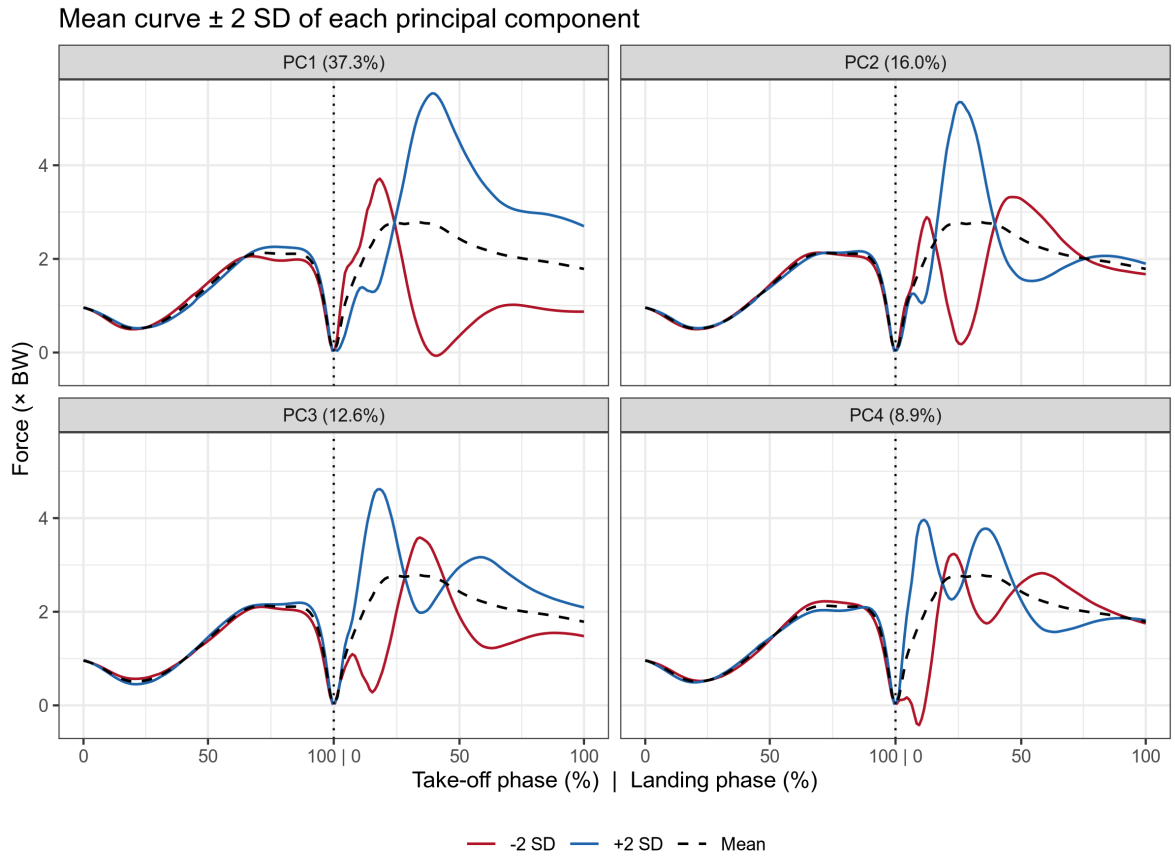


Figure 3. Mean curve (dashed) and mean ± 2 SD of each retained principal component (blue: +2 SD; red: −2 SD). The dotted line separates the take-off and landing phases.

- **PC1 (37.3%)** contrasts landings whose force peaks early in the landing phase (negative scores) with landings in which a large peak occurs later, around 40% of the phase, and force stays well above body weight until the lowest point (positive scores). It correlates most with landing peak force (r = 0.39) and only weakly with contraction time (r = -0.04) and countermovement depth (r = 0.01).
- **PC2–PC4** (16.0%, 12.6% and 8.9%) describe further differences in the timing and number of landing force peaks (single versus double peaks, earlier versus later). Each correlates with landing peak force (r = 0.22, 0.39, 0.45) and weakly with the take-off metrics (Section 4.5).

4.3 Do matched jumps have matched curves?

Table 5. Definition of the matched bands (45th–55th percentile of the matching metric).

Matching metric	Range	Trials	Athletes	Band mean ± SD
Jump height (cm)	26.03–28.24	82	51	27.10 ± 0.65
Concentric peak force (× BW)	2.25–2.32	81	58	2.28 ± 0.02

Table 6. Dispersion of component scores in the full sample and inside the matched bands. Ratio = SD in band / SD in full sample; 95% CI from 2,000 athlete-level bootstrap resamples.

Component	SD, full sample	SD, jump-height band	Jump-height band ratio (95% CI)	SD, rel. peak-force band	Rel. peak-force band ratio (95% CI)
PC1	7.28	6.45	0.89 (0.70–1.06)	7.33	1.01 (0.77–1.16)
PC2	4.77	4.62	0.97 (0.77–1.16)	5.01	1.05 (0.82–1.24)
PC3	4.22	4.03	0.95 (0.74–1.13)	4.25	1.00 (0.81–1.17)
PC4	3.56	4.16	1.17 (0.81–1.44)	3.75	1.05 (0.87–1.23)
Whole curve (√ total variance)	11.92	11.99	1.01	11.95	1.00

SD values of PC scores depend on the number of grid points (202) and are comparable only within this analysis; the ratios are scale-free.

Matching on jump height reduced the spread of PC1 only modestly (ratio 0.89, 95% CI 0.70–1.06, which includes 1) and left the other components and the whole curve essentially unchanged (ratios 0.97–1.17; whole curve 1.01). Matching on relative peak force had no visible effect (ratios 1.00–1.05). Figure 4 shows the 82 jump-height-matched curves: take-off curves look alike, while landing curves differ widely.

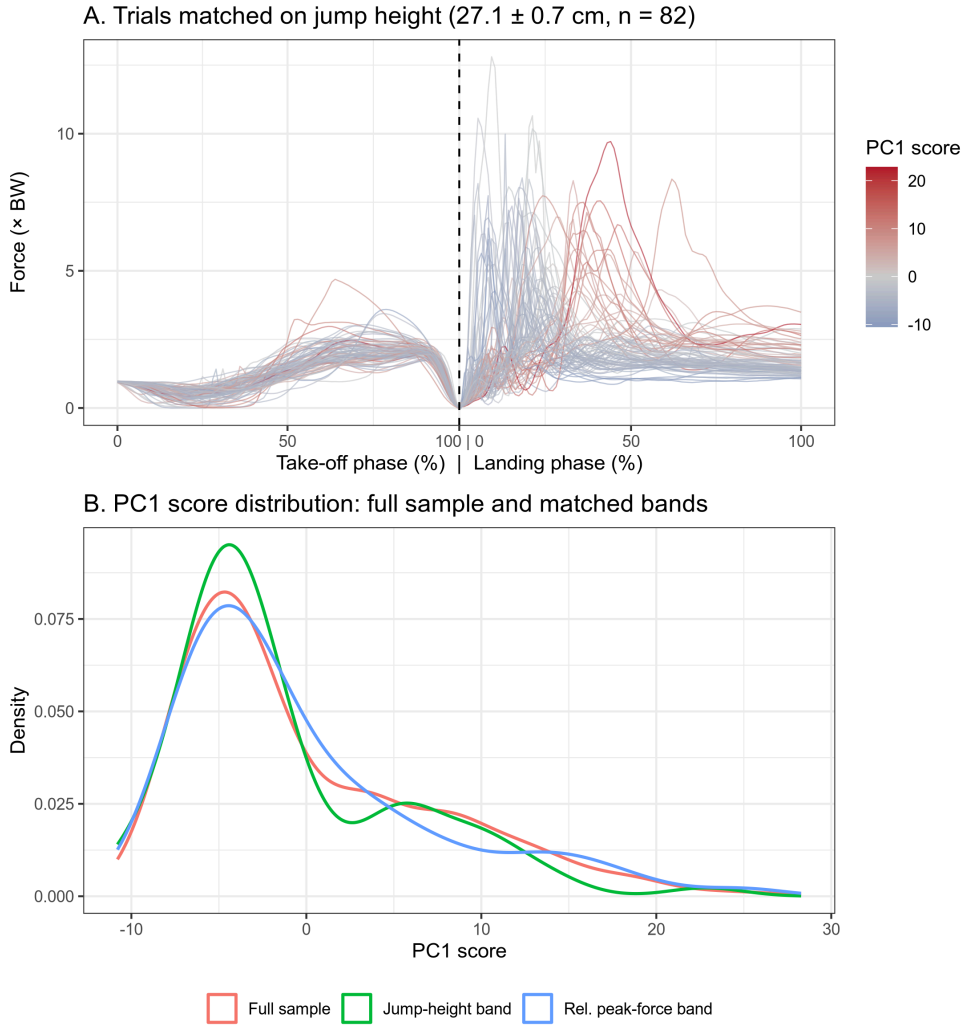


Figure 4. (A) Force–time curves of the 82 trials matched on jump height, coloured by PC1 score. (B) Distribution of PC1 scores in the full sample and in the two matched bands.

4.4 How much do conventional metrics explain?

Table 7. Variance of each component score explained by the four vendor metrics, age and body mass (OLS).

Component	Trials	R^2 (%)	Adj. R^2 (%)	Unexplained (%)
PC1	805	7.7	7.0	92.3
PC2	805	2.7	2.0	97.3
PC3	805	13.1	12.4	86.9
PC4	805	9.1	8.4	90.9

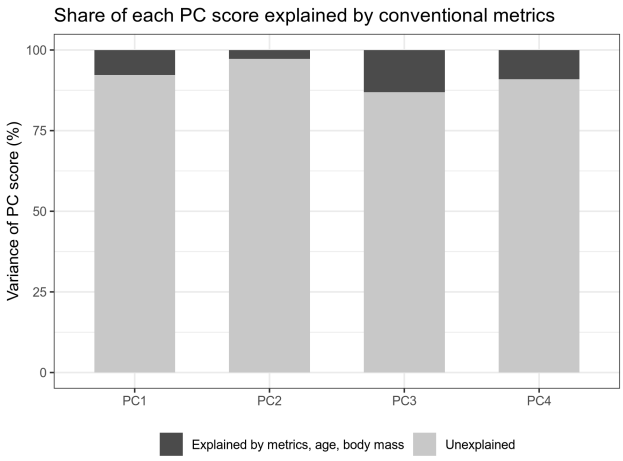


Figure 5. Share of each component score explained (dark) and unexplained (light) by the vendor metrics, age and body mass.

R^2 ranged from 2.7% to 13.1%, leaving 86.9–97.3% of each component unexplained. Regression coefficients for each model are in `S1_regression_coefficients_PC1..PC4.csv`.

4.5 Correlations with interpretable descriptors

Table 8. Pearson correlations between component scores and trial descriptors.

Descriptor	PC1	PC2	PC3	PC4
Landing peak force (× BW)	0.39	0.22	0.39	0.45
Contraction time (s)	-0.04	0.02	0.00	0.15
Countermovement depth (cm)	0.01	-0.00	-0.14	-0.18
Jump height (cm)	0.13	0.13	0.34	0.24
Concentric peak force (N)	0.10	0.02	0.20	0.03
Concentric peak force (× BW)	0.14	-0.02	0.09	-0.09
Eccentric braking impulse (N s)	0.04	-0.00	0.20	0.05
Concentric impulse 100 ms (N s)	0.15	0.00	0.19	-0.01
Age (years)	0.14	0.08	0.26	0.13
Body mass (kg)	0.02	0.03	0.18	0.09

Landing peak force is the descriptor most strongly associated with every retained component ($r = 0.22$ to 0.45). The four vendor metrics show weak correlations ($|r| \leq 0.34$), and take-off timing and countermovement depth are nearly unrelated to the components ($|r| \leq 0.18$).

5. Robustness

5.1 Alternative landing window: 1 s after touch-down

The main analysis ends the landing at the lowest centre-of-mass point. The alternative uses the vendor's trial window, which ends 1 s after touch-down. In that version the landing impact occupies only the first few percent of the landing block, the last part is quiet standing, and the take-off block gains relative weight. The results change in an important way (Table 9, Figure 6):

- Four components explain 62.2% instead of 74.8%.
- PC3 becomes a **take-off component** (96.3% of its loading in take-off), strongly related to contraction time (r = -0.79) and relative peak force (r = 0.77). It is **68.3% predictable** from the vendor metrics, age and body mass (66.4% under athlete-grouped cross-validation).
- The landing components (PC1, PC2, PC4) remain poorly explained (R^2 19.1%, 5.3%, 17.8%), and PC1 is again tied to landing peak force (r = 0.71).

The honest reading is that the "largely unexplained" conclusion applies to **landing** shape. Take-off shape, when it forms its own component, is substantially captured by the conventional metrics. Which landing definition is used should be stated explicitly in the paper.

Table 9. Main analysis versus the 1-s landing sensitivity analysis.

Quantity	Main: lowest-COM landing	Sensitivity: 1-s landing
Trials / athletes after QC	805 / 212	811 / 212
Variance explained, PC1–PC4 (%)	37.3, 16.0, 12.6, 8.9	20.2, 16.6, 15.1, 10.2
Cumulative, first 4 PCs (%)	74.8	62.2
Landing share of loading, PC1–PC4 (%)	99.7, 99.9, 99.5, 99.3	96.4, 92.5, 3.7, 80.2
Jump-height band SD ratio, PC1 (95% CI)	0.89 (0.70–1.06)	0.82 (0.69–0.96)
Whole-curve SD ratio, jump-height band	1.01	1.01
R^2, PC1–PC4 (%)	7.7, 2.7, 13.1, 9.1	19.1, 5.3, 68.3, 17.8
Athlete-CV R^2, PC1–PC4 (%)	4.0, 0.3, 11.8, 6.3	13.4, 2.8, 66.4, 15.1
r (PC1, landing peak force)	0.39	0.71
r (PC3, contraction time)	0.00	-0.79

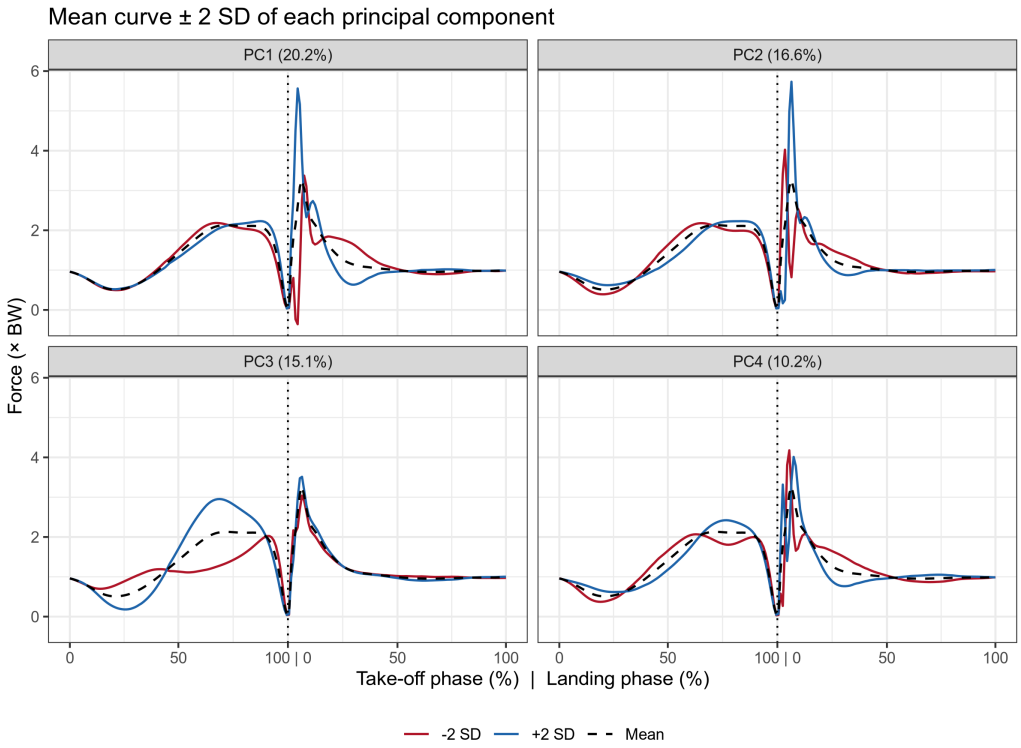


Figure 6. Sensitivity analysis (1-s landing window): mean ± 2 SD of each retained component. PC3 is a take-off timing component.

5.2 Athlete-grouped cross-validation and repeated trials

In-sample R^2 can be optimistic, and athletes contribute several trials (3.8 on average). Table 10 compares the in-sample R^2 with (a) 5-fold cross-validation in which athletes, not trials, were split into folds; (b) one randomly chosen trial per athlete (seed 2026); and (c) athlete means. Out-of-sample R^2 was lower (0.3–11.8%), and one trial per athlete or athlete means gave 4.3–21.2%, so the conclusion does not depend on repeated trials.

Table 10. Variance explained under alternative evaluation schemes (main analysis).

Component	R^2 (%)	Athlete-CV R^2 (%)	One trial/athlete R^2 (%)	Athlete means R^2 (%)
PC1	7.7	4.0	10.0	11.1
PC2	2.7	0.3	4.3	5.4
PC3	13.1	11.8	17.8	21.2
PC4	9.1	6.3	11.1	13.8

5.3 Width of the matched band

Table 11. PC score SD ratio (band / full sample) for different jump-height band definitions (main analysis).

Band definition	Trials	Jump height mean ± SD (cm)	PC1	PC2	PC3	PC4
45th-55th percentile (main)	82	27.10 ± 0.65	0.89	0.97	0.95	1.17
40th-60th percentile	161	27.07 ± 1.38	0.87	1.02	0.84	1.07
Median ± 1 cm	70	26.92 ± 0.53	0.88	0.86	0.97	1.17
Median ± 2 cm	133	27.05 ± 1.12	0.88	0.97	0.86	1.06

PC1 ratios stayed between 0.87 and 0.89 across band definitions, so the result does not depend on the chosen band width.

6. Comparison with the submitted abstract

The abstract (submitted 26 September 2026) was based on an earlier analysis whose code and intermediate files could not be located. The analysis was therefore rebuilt from the raw ForceDecks data following the abstract's stated method. The direction of every conclusion is reproduced, but several numbers differ (Table 12).

Table 12. Key numbers in the submitted abstract and in this report.

Quantity	Submitted abstract	This report (main)	1-s landing
Trials / athletes after QC	817 / 213	805 / 212	811 / 212
Variance, first 4 PCs (%)	71	74.8	62.2
Jump-height band	25.9 ± 1.0 cm, n = 71	27.1 ± 0.7 cm, n = 82	27.1 ± 0.7 cm, n = 82
PC1 SD, band vs full sample	5.21 vs 5.25	6.45 vs 7.28	2.11 vs 2.56
PC1 SD ratio (band / full)	≈0.99	0.89	0.82
R^2 of PC scores (range, %)	5–17	2.7–13.1	5.3–68.3
Unexplained (range, %)	83–95	86.9–97.3	31.7–94.7
Main correlate of PC1	Landing peak force	Landing peak force (r = 0.39)	Landing peak force (r = 0.71)

- **Sample.** 805 trials from 212 athletes versus 817 from 213. The abstract's QC rules are not documented; the rules here are listed in Table 1.
- **Variance explained.** 74.8% versus 71%. The abstract does not state where the landing phase ends; the value depends strongly on this choice (62.2% with a 1-s window).
- **Matched-band SD ratio.** 0.89 versus about 0.99 for PC1. The confidence interval (0.70–1.06) includes 1, so the abstract's qualitative claim holds, but matching on jump height does reduce PC1 spread somewhat. The absolute SDs (5.21 vs 5.25 in the abstract) cannot be compared because they depend on the number of grid points used. The band itself also differs (the abstract's band was 25.9 ± 1.0 cm, n = 71).
- **R^2.** 2.7–13.1% versus 5–17%: consistent in range and conclusion for the main analysis, but not for the 1-s landing version (Section 5.1).

The full paper should report the numbers in this report, not the abstract's numbers.

7. Practical implications

- **Same jump height ≠ same jump.** Two players with the same jump height can land very differently. Ranking or grouping players by jump height alone hides this.
- **Look at the landing.** Most curve-shape variation, and the part least explained by standard metrics, lies in how players absorb the landing – how large the force peak is and when it occurs. Landing metrics and the landing curve deserve a place in routine reports alongside take-off metrics.
- **Use data already collected.** The curves come from routine CMJ testing; no new protocol or equipment is needed. A component score can be computed for each new trial and tracked over time for each player.
- **Treat take-off metrics as adequate for take-off.** The sensitivity analysis shows that take-off timing is largely captured by conventional metrics; the added value of curve analysis lies mainly in landing.

- **Be careful with interpretation.** Landing differences could reflect strategy, fatigue, instruction or simply how a player was told to land. They should prompt a closer look, not a diagnosis.

8. Limitations

- **Landing definition.** Results depend on where the landing phase is taken to end (Section 5.1). The lowest-centre-of-mass end point relies on body mass and on the velocity estimated from flight time; it could not be found for 6 trials.
- **Landing instructions not standardised.** Routine tests do not control how athletes land, so part of the landing variation may reflect instruction or habit rather than neuromuscular capacity.
- **Repeated trials.** The main analysis treats trials as independent. Athlete-level checks (cross-validation, one trial per athlete, athlete means, athlete bootstrap) give the same conclusions, but a mixed-effects model would separate within- and between-athlete variation formally.
- **Population.** Male football players from one centre, aged 8.8–37.9 years at first test (median 14.1); results may not generalise to other sports, women or other age groups.
- **Linear PCA and simple alignment.** PCA on time-normalised curves treats timing and amplitude differences together. Registration of further landmarks (for example the landing force peak) or functional PCA could separate them.
- **Abstract numbers not reproduced exactly.** The original analysis files were not available (Section 6).
- **Linear regression only.** R^2 measures linear association; non-linear models could explain more, although the weak correlations suggest the gain would be limited.

9. Reproducibility

All code, the de-identified data and every output are in the local Git repository `baiconf2026-jump-pca` (MIT licence). The analysis uses R 4.4.3 with dplyr, tidyr, ggplot2 and patchwork; exact versions are in `packages.csv` and `sessionInfo.txt`. The only random steps (cross-validation folds, one-trial selection, bootstrap) use seed 2026.

`data/`	De-identified trial file (833 trials, before QC) and the analysis-ready body-weight-normalised concatenated curves
`R/00_extract_from_raw.R`	Private step: builds the trial file from raw VALD exports (not run by run_all.R)
`R/01 … R/08`	Load and QC · normalise and concatenate · PCA · matched bands · regression · perturbation figures and correlations · tables · robustness
`outputs_lowest_com/`	Main analysis: figures (PNG 300 dpi, PDF) and tables (CSV)
`outputs/`	Sensitivity analysis with the 1-s landing window
`report/`	This report and its source (build_report.py)

```
Rscript install_packages.R                    # once
LANDING_WINDOW=lowest_com Rscript run_all.R   # main analysis   -> outputs_lowest_com/
Rscript run_all.R                             # 1-s sensitivity -> outputs/
python3 report/build_report.py                # this PDF
```

References

1. Cormie, P., McBride, J. M., & McCaulley, G. O. (2009). Power-time, force-time, and velocity-time curve analysis of the countermovement jump: Impact of training. *Journal of Strength and Conditioning Research, 23*(1), 177–186. https://doi.org/10.1519/JSC.0b013e3181889324
2. Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall/CRC. https://doi.org/10.1007/978-1-4899-4541-9
3. Harry, J. R., Paquette, M. R., Schilling, B. K., Barker, L. A., James, C. R., & Dufek, J. S. (2018). Kinetic and electromyographic subphase characteristics with relation to countermovement vertical jump performance. *Journal of Applied Biomechanics, 34*(4), 291–297. https://doi.org/10.1123/jab.2017-0305
4. Harry, J. R., Barker, L. A., & Paquette, M. R. (2020). A joint power approach to define countermovement jump phases using force platforms. *Medicine & Science in Sports & Exercise, 52*(4), 993–1000. https://doi.org/10.1249/MSS.0000000000002197
5. Hite, M. D., Simms, A. J., Stewart, M. J., Chowning, L. D., & Harry, J. R. (2025). Differences in countermovement jump landing characteristics and joint works across levels of perceived effort. *Journal of Biomechanics, 184*, 112645. https://doi.org/10.1016/j.jbiomech.2025.112645
6. Jolliffe, I. T. (2002). *Principal component analysis* (2nd ed.). Springer. https://doi.org/10.1007/b98835
7. Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A, 374*(2065), 20150202. https://doi.org/10.1098/rsta.2015.0202
8. Linthorne, N. P. (2001). Analysis of standing vertical jumps using a force platform. *American Journal of Physics, 69*(11), 1198–1204. https://doi.org/10.1119/1.1397460
9. McMahon, J. J., Suchomel, T. J., Lake, J. P., & Comfort, P. (2018). Understanding the key phases of the countermovement jump force-time curve. *Strength and Conditioning Journal, 40*(4), 96–106. https://doi.org/10.1519/SSC.0000000000000375
10. Owen, N. J., Watkins, J., Kilduff, L. P., Bevan, H. R., & Bennett, M. A. (2014). Development of a criterion method to determine peak mechanical power output in a countermovement jump. *Journal of Strength and Conditioning Research, 28*(6), 1552–1558. https://doi.org/10.1519/JSC.0000000000000311

11. Ramsay, J. O., & Silverman, B. W. (2005). *Functional data analysis* (2nd ed.). Springer. https://doi.org/10.1007/b98888
12. Richter, C., O'Connor, N. E., Marshall, B., & Moran, K. (2014a). Analysis of characterizing phases on waveforms: An application to vertical jumps. *Journal of Applied Biomechanics, 30*(2), 316–321. https://doi.org/10.1123/jab.2012-0218
13. Richter, C., O'Connor, N. E., Marshall, B., & Moran, K. (2014b). Comparison of discrete-point vs. dimensionality-reduction techniques for describing performance-related aspects of maximal vertical jumping. *Journal of Biomechanics, 47*(12), 3012–3017. https://doi.org/10.1016/j.jbiomech.2014.07.001
14. Warmenhoven, J., Cobley, S., Draper, C., Harrison, A., Bargary, N., & Smith, R. (2019). Considerations for the use of functional principal components analysis in sports biomechanics: Examples from on-water rowing. *Sports Biomechanics, 18*(3), 317–341. https://doi.org/10.1080/14763141.2017.1392594